

programming. Microcontrollers and control programming is exclusively an electronics engineer's domain. Data acquisition systems or control systems with real-time operating systems also require the hands of an electronics engineer. Only an electronics engineer is equipped to understand how data travels from one digital system to another. A non-programming area where electronics engineers get involved is board design.

"DSP chips is another area where electronics engineers can play a significant role, as they have an added advantage over software engineers in writing software algorithms for hardware codecs," Verma adds.

These jobs require device driver and hardware related knowledge, and a good understanding of the system on chip. The programming language is 'C' and Assembly. Electronics engineers play a very important role in the development of embedded systems, network switching and handheld devices. They can take entrepreneurial path too using Open Source.

Build your Open Source skills

Open Source electronics is not a common topic found in university syllabi. Instead, most people refer to resources on the Internet. Luke Soules, the founder of ifixit.com, feels that those without hardware knowledge can easily get started with electronics and those who have not programmed before can experiment on boards with Open Source projects.

"The best example of a successful Open Source hardware project is Arduino. It is a microcontroller platform with digital input and output, and analogue input and output that

Recommended readings

- 'Embedded Linux Primer: A Practical Real-World Approach' by Christopher Hallinan
- 'Embedded Linux System Design and Development' by P. Raghavan, Amol Lad, Sriram Neelakandan
- 'Building Embedded Linux Systems' by Karim Yaghmour
- 'Linux for Embedded and Real-Time Applications, Second Edition' by Doug Abbott
- 'Embedded Linux: Hardware, Software, and Interfacing' by Dr Craig Hollibaugh
- 'Embedded Linux' by John Lombardo
- 'The Linux TCP/IP Stack: Networking for Embedded Systems' by Thomas Herbert
- 'Hacking the Tivo' by William von Hagen
- 'Practical Linux Programming: Device Drivers, Embedded systems, and the Internet' by Ashfaq A. Khan

lets you build things," Soules stated in a panel discussion titled 'The Rise of Profitable Open Source Hardware' available online.

Arduino is an Open Source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for artists, designers, hobbyists and anyone interested in creating interactive objects or environments.

Arduino can sense the environment by receiving input from a variety of sensors and affect its surroundings by controlling lights, motors and other actuators. The microcontroller on the board is programmed using the Arduino programming language based on Wiring and the Arduino development environment based on Processing. Arduino projects can be standalone or they can communicate with the software on running on a computer (Processing, MATLAB and MAX/MSP).

Vinay Chaddha, an innovator and embedded software and hardware developer, feels most colleges don't teach Open Source electronics due to lack of awareness. In such a situation, reading electronics periodicals and blogs can help in understanding electronics.

Chaddha says, "Open Source elec-

tronics is simple, everyday electronics. The only addition is that all the information is available in public domain and there are many people working on it. So there is a large community and support is easy. Also, you do not have to start from scratch."

He recommends that doing internships and experimenting is much more beneficial than undergoing paid training. A good way to start is exploring community projects, and working and tweaking existing device drivers that are a part of Open Source community.

"You can buy one of the boards such as Beagleboard (which costs around \$100), use the supporting material that is available on the website and build your own applications. Hands-on implementation is the key to learning. We tend to look for a strong embedded systems background with good knowledge of Linux and 'C' as the candidates have to handle product application and development," reveals Giri Krishna, CEO of Silvan Innovation Labs.

Krishna adds, "We work with the development boards and kits provided by vendors to develop our own applications. For example, we have used the Leopard board to develop IP surveillance cameras."

Meanwhile, Kwang feels that Open Source does not present any barrier to entry. "One's willingness to embrace Open Source and a keen eye for opportunities, changes and demands are the most important factors. Linux is fundamentally a great place to start. SourceForge.com is another excellent

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